

H2 — ML + Embedded Track, Week 1

Member: _____

Project setup (do first)

1. Fork the repo: <https://github.com/slvDev/esp32-ai>
2. Clone your fork. Add the original as upstream.
3. Read README.md, RESULTS.md, and research/tinystories/README.md.
4. Write 5 lines in the shared Drive doc: what you think the project does and where the hard part is.

Your task this week

Goal: the pipeline from model file to flashed board is documented.

Steps: 1. Read scripts/fetch_model.sh, scripts/deploy.sh, scripts/benchmark_device.py. 2. Trace the SHA-256 pin logic in fetch_model.sh. Note what must match before install. 3. Write the pipeline doc: steps, diagram, failure points.

Deliverable: the pipeline doc as a markdown file in the club repo.

Verify: H4 follows the doc and can name the three stages of deploy.sh.

Learning this week

Neural network basics. Watch 3Blue1Brown episode 1. Write the MLP from the video in numpy.

Resources

- scripts/: <https://github.com/slvDev/esp32-ai/tree/main/scripts>
- 3Blue1Brown neural networks: https://www.youtube.com/playlist?list=PLZHQObOWTQDNU6R1_67000Dx_ZCJB-3pi

Done when

- [] Pipeline doc exists in the club repo
- [] You can say what the project does in one sentence